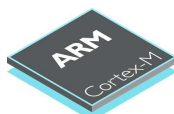


ARM Cortex-M4 Processors Features
The Cortex-M Processor Families
ARM Architectures and Instruction Sets
Cortex-M4 Block Diagram
WIC
NVIC
Debug and Trace Systems
Bus Systems
Inside the ARM Processor



1

ARM Cortex-M Processor Families

ARM Cortex-A family:

- Applications processors
- Support OS and high-performance applications
- Such as Smartphones, Smart TV



ARM Cortex-R family:

- Real-time processors with high performance and high reliability
- Support real-time processing and mission-critical control



ARM Cortex-M family:

- Microcontroller
- Cost-sensitive, support SoC



2

What is Cortex Processor !

- The new generation of the ARM Processor
- Consists of ARM CPU + System Peripherals
- Thumb-2 Technology
- Faster, more predictable interrupts and better debugging support
- On Chip Bus Interface based on ARM AMBA
- Nested Vector Interrupt Controller
- Unaligned memory access

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Series of Cortex Processor

- Cortex-A: General Purpose High end applications, System Clock > 1GHz
- Cortex-R: Real-Time mid end applications, 600MHz > System Clock < 400MHz
- Cortex-M: Special Purpose low end applications, System Clock < 200MHz

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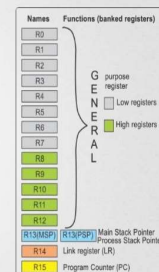
Cortex-M Processors

- The First of the Cortex generation, started with Cortex-M3 in 2005
- Cortex-M3 & Cortex-M4 are 32-bit Architecture
- Thumb-2 Instruction set
- Three Stage pipeline design
- Harvard Architecture with unified memory space
- Cortex-M4 Supports: SIMD, MAC, Float point instructions (single precision), Saturating arithmetic instructions
- Usage: Microcontrollers Cores

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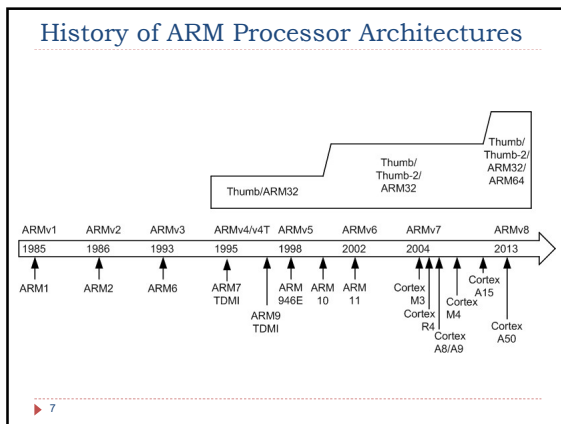
Cortex-M CPU

- 32-bit RISC
- 16 registers
- 3 stages pipeline [fetch, decode, execute]
- Branch prediction
- Load-store architecture

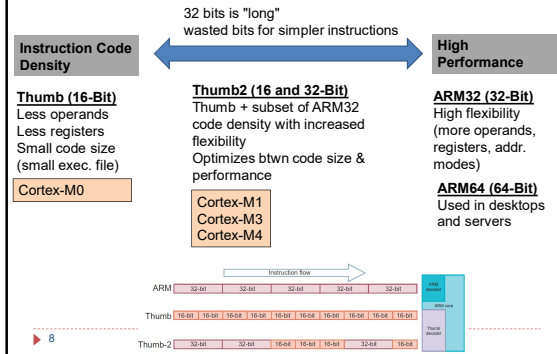


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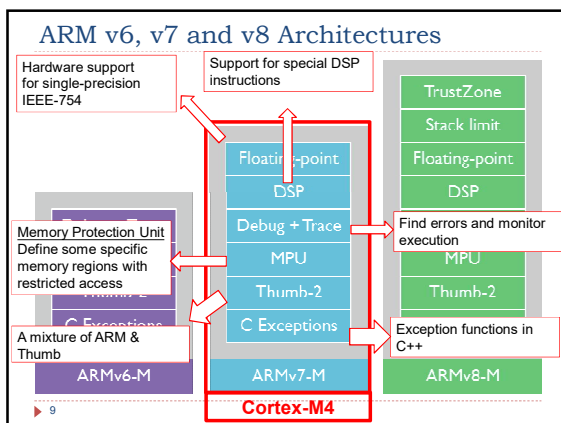
History of ARM Processor Architectures



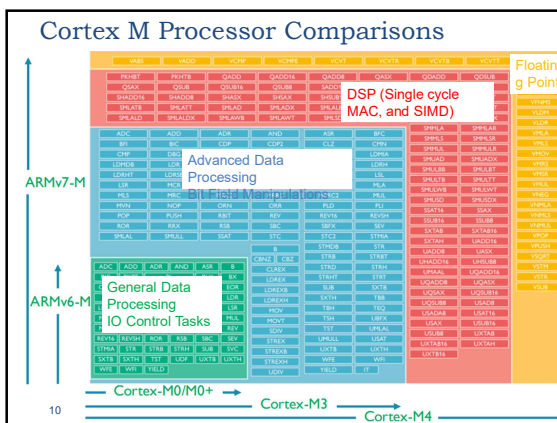
ARM Instruction Sets



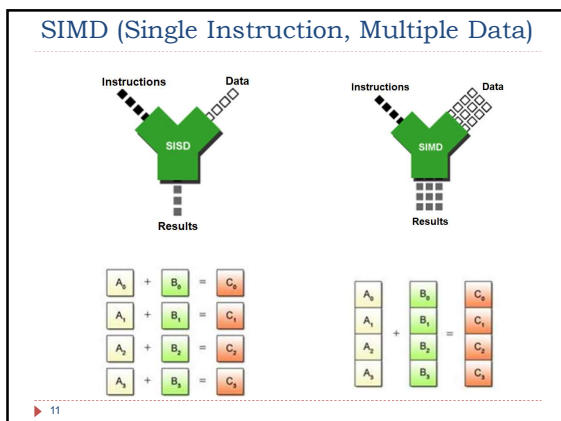
ARM v6, v7 and v8 Architectures



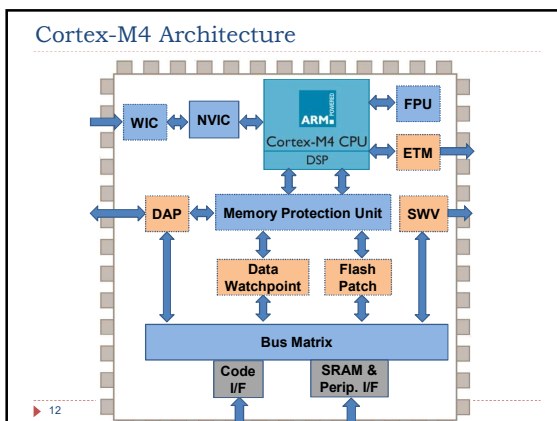
Cortex M Processor Comparisons



SIMD (Single Instruction, Multiple Data)

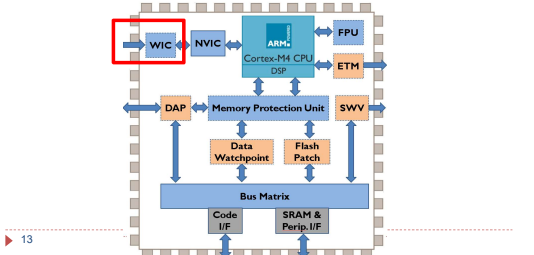


Cortex-M4 Architecture



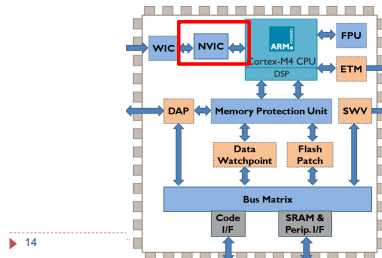
WIC (Wake-Up Interrupt Controller)

- Enables ultra low-power standby operation
 - sleep mode by shutting down most of the components.
- When an interrupt request is detected,
 - inform the power management unit to power up the system.



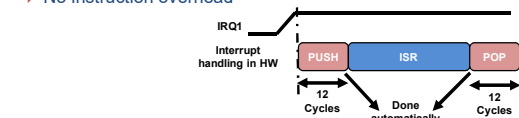
NVIC (Nested Vectored Interrupt Controller)

- Fast execution of interrupts
 - Lower latency
- Deterministic response
 - Reaches ISR in 12 cycles

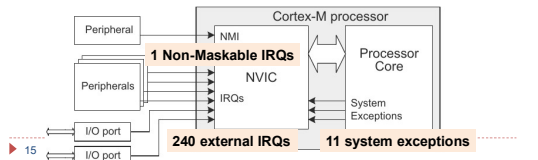


NVIC (Nested Vectored Interrupt Controller)

- State automatically saved/restored on ISR entry/exit
 - No instruction overhead



- Programmable priority levels



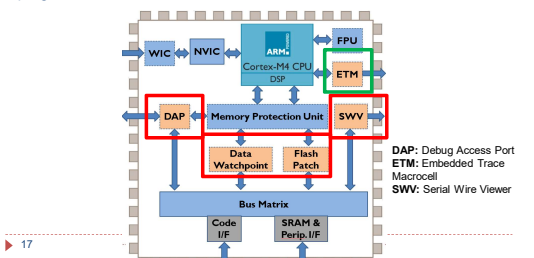
NVIC Vector Table

- Gets the ISR addresses from the vector table.

Exception Type	Vector Number	Priority	Vector Address or Offset
-	0	-	0x0000.0000
Reset	1	-3 (highest)	0x0000.0004
Non-Maskable Interrupt (NMI)	2	-2	0x0000.0008
Hard Fault	3	-1	0x0000.000C
Memory Management	4	programmable	0x0000.0010
Bus Fault	5	programmable	0x0000.0014
Usage Fault	6	programmable	0x0000.0018
-	7-10	-	-
SVCall	11	programmable	0x0000.002C
Debug Monitor	12	programmable	0x0000.0030
-	13	-	-
PendSV	14	programmable	0x0000.0038
SysTick	15	programmable	0x0000.003C
Interrupts	16 and above	programmable	0x0000.0040 and above

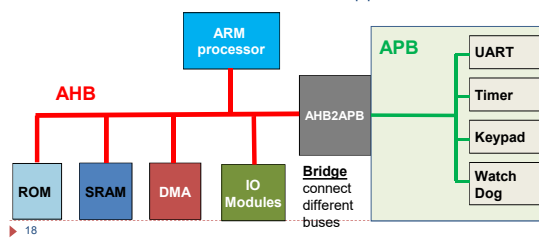
Debug and Trace System

- Debug**
 - Examine memory and registers
 - Insert breakpoints
 - Halt the processor
 - Run or Single-step through programs
- Trace**
 - Getting information regarding program's execution
 - Used for testing and optimization



Bus Systems

- Advanced High Performance Bus**
 - Main System Bus
 - High performance
 - Pipelined operation
 - Burst transfers
 - Multiple bus masters
 - Split transactions
- Advanced Peripheral Bus**
 - For low speed peripherals
 - Low power
 - Simple interface
 - No bursts
 - No pipelined transfers



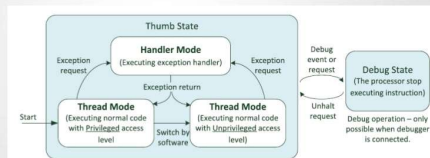
Cortex-M System buses

- ICode bus
Fetch opcode from ROM
- DCode bus
Read constant data from ROM
- System bus
Read/Write data from RAM or I/O, fetch opcode from RAM
- PPB
Read/Write data from internal peripherals
- like the NVIC
- AHB
Read/Write data from high-speed I/O and parallel ports (**M4 Only**)

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CPU operating modes

- Privileged access level [use MSP as the stack pointer]
- Unprivileged access level [use PSP as the stack pointer]



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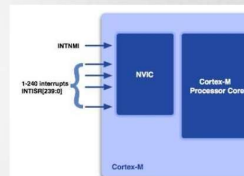
Power Modes

- WFI
• Puts the CPU into sleep mode until an interrupt happen
- WFE
• Put the CPU into sleep mode until an external event happen
- LEEPDEEP
• CPU Enters the deep sleep mode
- Halts Peripherals, Processor Clock, flash memory and PLL
- SLEEPONEXIT
• Puts CPU into sleep mode after finishing an ISR
- SEVONPEND
• An interrupt can generate an event even if disabled
- The interrupt status needs to be 0 before entering WFE

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Nested Vectored Interrupt Controller

- Flexible exception and interrupt management
- Nested exception / interrupt support
- Vectored exception / interrupt entry
- Interrupt masking
- Supports 240 interrupt requests
- 8 to 256 interrupt priority levels
- Configurable by the manufacturers to the needed number of interrupts



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System Tick Timer (SysTick)

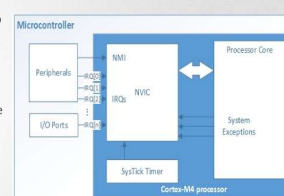
- A 24-bit Down Counter
- Auto Acknowledge
- Auto reload
- Consists of 4 registers :

Name	Address
SysTick Control and Status	0xE000E010
SysTick Reload value	0xE000E014
SysTick Current value	0xE000E018
SysTick Calibration value [optional]	0xE000E01C

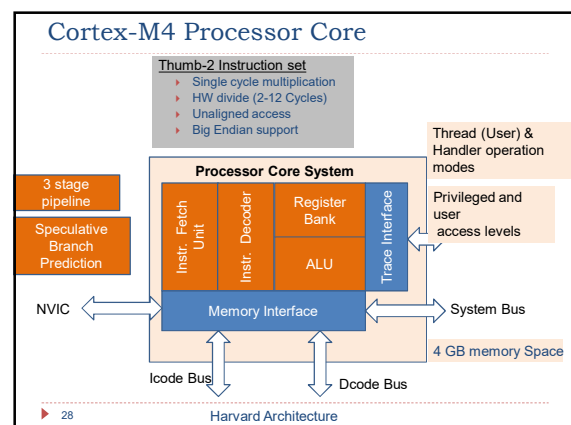
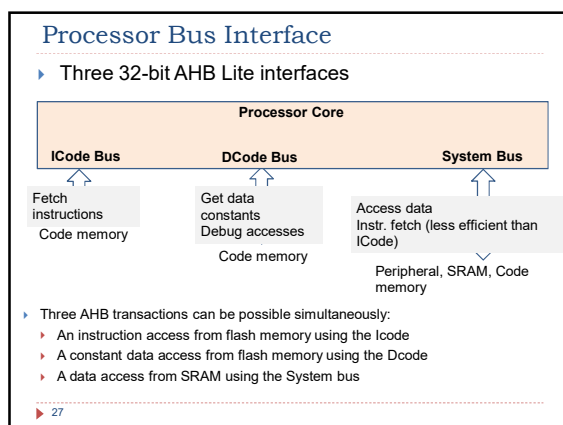
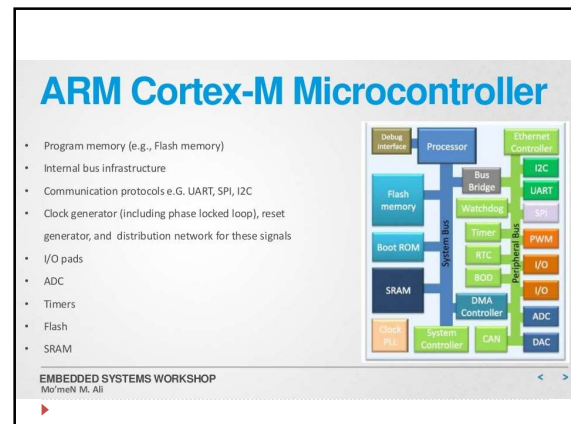
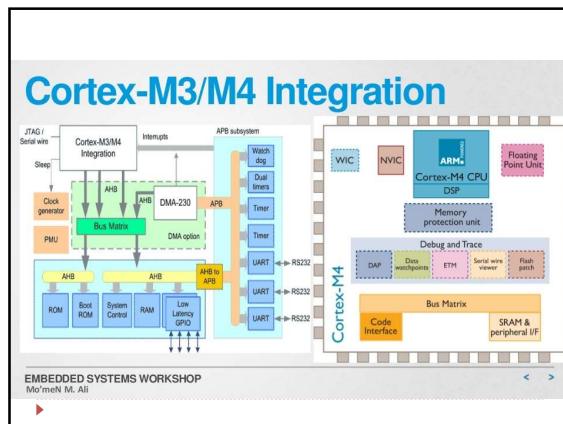
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Working sequence of the NVIC processor

1. A peripheral asserts an interrupt request to the processor
2. The processor suspends the currently executing task
3. The processor executes an interrupt service routine (ISR) to service the peripheral, and optionally clear the interrupt request by software if needed.
4. The processor resumes the previously suspended task.



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Area and Power of Cortex-M4 Processor Core

- ▶ Low dynamic power
- ▶ Small area

Process (nm)	Dynamic Power (μW/MHz)	Area (mm ²)
180 nm	157	0.56
90 nm	33	0.17
40 nm	8	0.04

0.04 mm² = 0.2mm x 0.2mm
Hair thickness ≈ 0.1mm

▶ 29

